



Indices and scientific notation

Teacher resource

Question 1

Which of the following statements is true?

☐ $3^4 = 4^3$

☐ $3^4 = 6^2$

☐ $2^6 = 6^2$

☐ $2^6 = 4^3$

Answer: D

Skill: Students interpret index notation.

Background information

Students should be able to read, write, say and interpret mathematical statements in symbolic form. It is important for students to connect patterns expressed using the reduced symbolic form of index notation with growth patterns produced by repeated multiplication. For example:

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

Students should recognise this as a doubling pattern, an example of a multiplicative pattern, with the terms of the sequence 'getting bigger by an increasing amount'. This is in contrast to an additive pattern in which the terms 'get bigger by a constant amount'.

Additional questions

1. What is the value of 8×4^2 ? How does this differ from $(8 \times 4)^2$?
2. What is the value of $(8 + 4)^2$?

Question 2

$$4^4 - 4^2$$

What is the value of this expression?

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| 8 | 16 | 32 | 240 |
| ☐ | ☐ | ☐ | ☐ |

Answer: D

Skill: Students evaluate an algebraic expression involving the use of indices.

Additional questions

Determine the value of these expressions:

1. $(4 - 2)^4$
2. $(4^2 - 2^2)^2$
3. $2^4 - 4^2$

Question 3

$$a^3 = b^2$$

Which of the following would **not** make this statement true?

- ☐ a = 1 and b = 1
- ☐ a = 1 and b = -1
- ☐ a = 2 and b = 3
- ☐ a = 4 and b = 8

Answer: C

Skill: Students apply knowledge of index notation.

Additional questions

1. What is the value of $(-6)^2$? How does this compare to the value of 6^{-2} ?
2. What is the value of $-(6^2)$? How does this compare to the value of $(-6)^{-2}$?

Question 4

A bacterium has a mass of 6.8×10^{-13} grams. Another bacterium has a mass of 5.4×10^{-12} grams. What is the total mass in grams of the two bacteria?

- ☐ 6.08×10^{-13}
- ☐ 6.08×10^{-12}
- ☐ 1.22×10^{-13}
- ☐ 1.22×10^{-12}

Answer: B

Skill: Students interpret and use scientific notation.

Additional questions

1. A bacterium has a mass of 6.8×10^{-13} grams. A second bacterium has a mass that is twice the mass of this bacterium. What is the mass of the second bacterium?
2. A bacterium has a mass of 6.8×10^{-13} grams. A second bacterium has a mass that is tens times smaller. What is the mass of the second bacterium?

Student activity: Indices and scientific notation

Question 1

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- ☐ $3^4 = 6^2$
- ☐ $2^6 = 6^2$
- ☐ $2^6 = 4^3$

Question 3

$$a^3 = b^2$$

Which of the following would **not** make this statement true?

- ☐ $a = 1$ and $b = 1$
- ☐ $a = 1$ and $b = -1$
- ☐ $a = 2$ and $b = 3$
- ☐ $a = 4$ and $b = 8$

Question 2

$$4^4 - 4^2$$

What is the value of this expression?

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| 8 | 16 | 32 | 240 |
| ☐ | ☐ | ☐ | ☐ |

Question 4

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